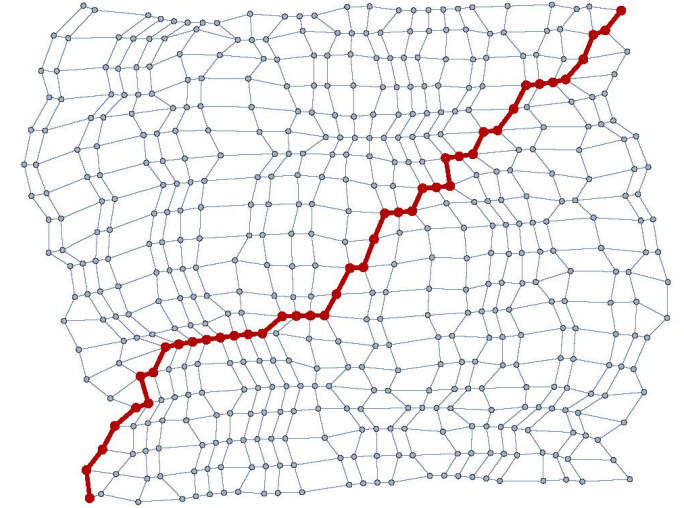




Aspects of KPZ universality: random metrics & growth processes

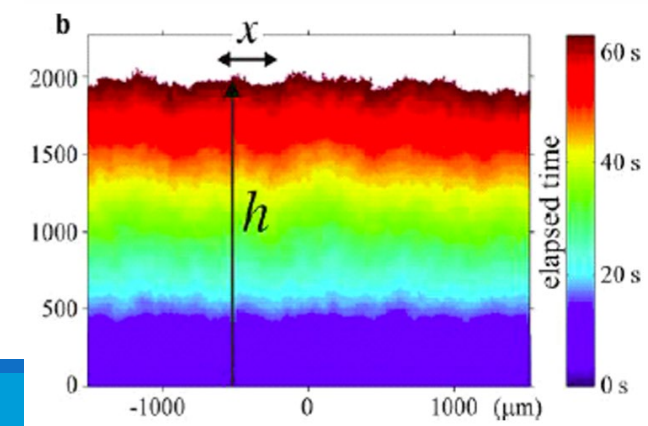
Lingfu Zhang

California Institute of Technology

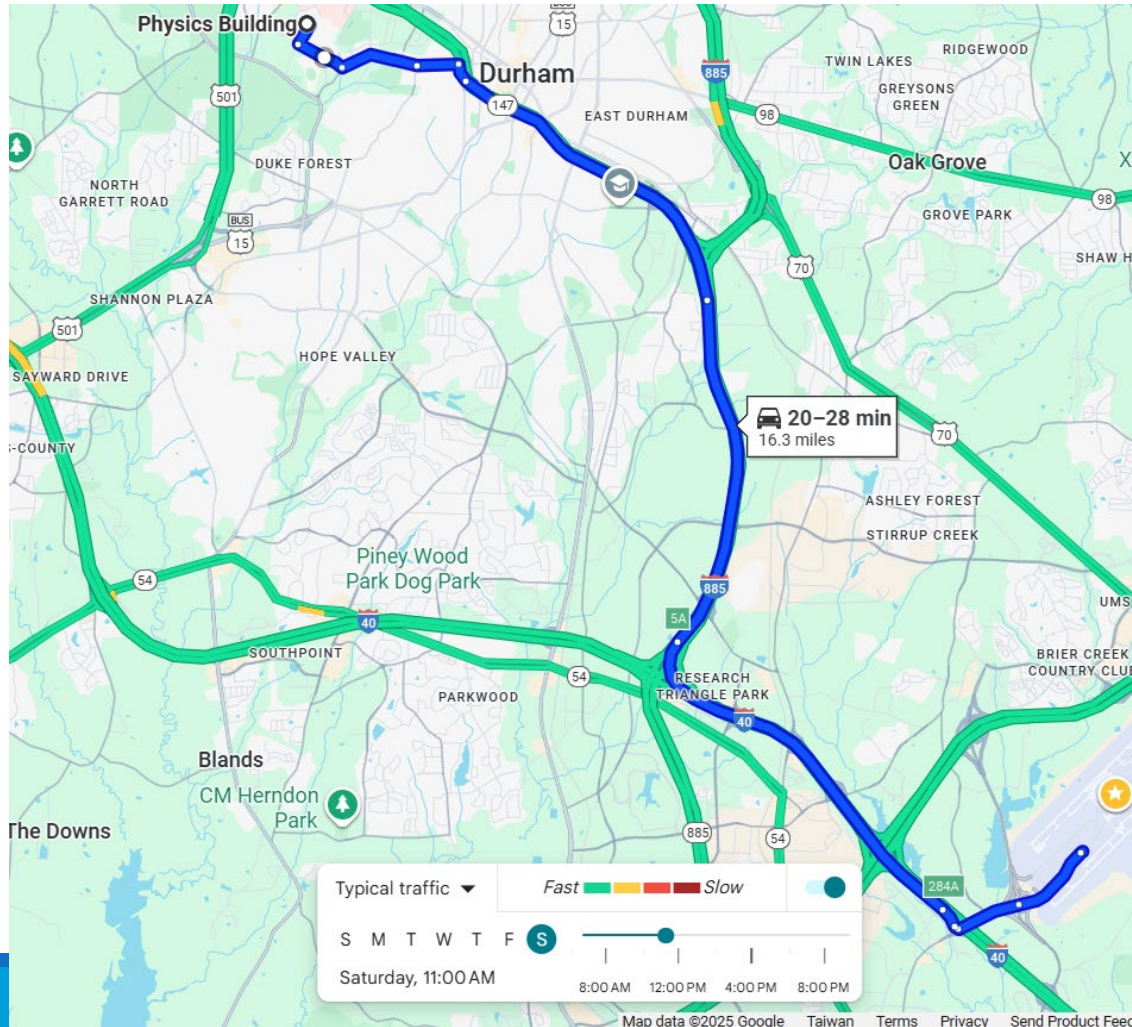
Triangle Probability Forum

Sep 12, 2025

Joint work with Duncan Dauvergne (arxiv:2412.13032)



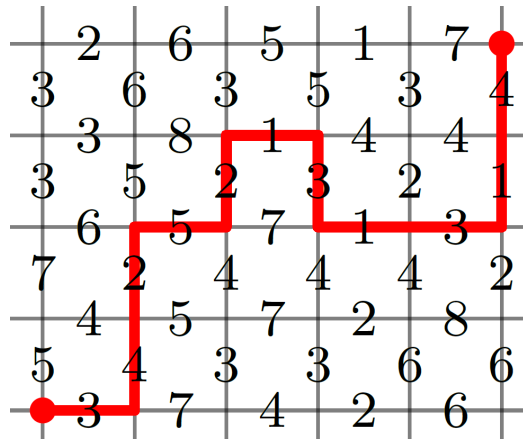
Planar random metric



Say I am planning my trip to the airport on Saturday morning.

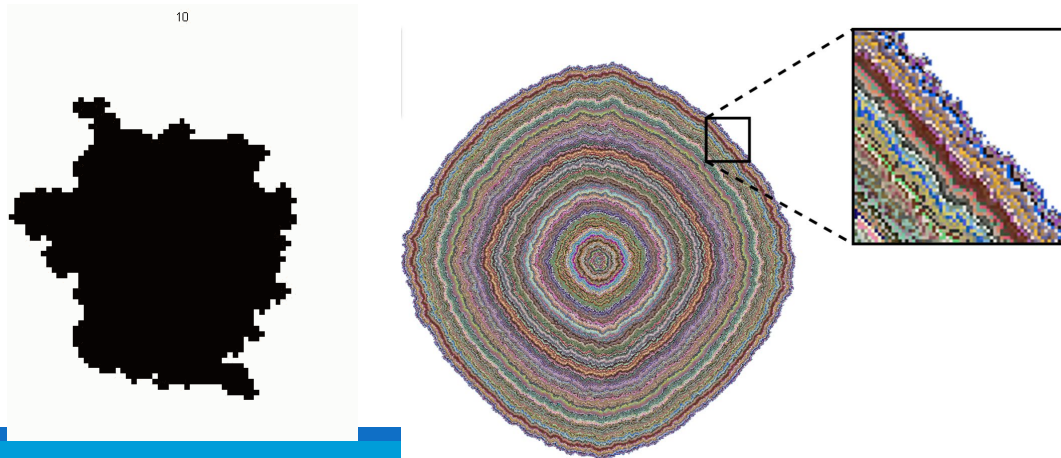
- ❖ How long does it take?
- ❖ How confident are you about that? (i.e., fluctuation?)
- ❖ What the route is like?

First Passage Percolation (FPP)



- i.i.d. random on each edge
- distance between two points:
minimum sum over all paths

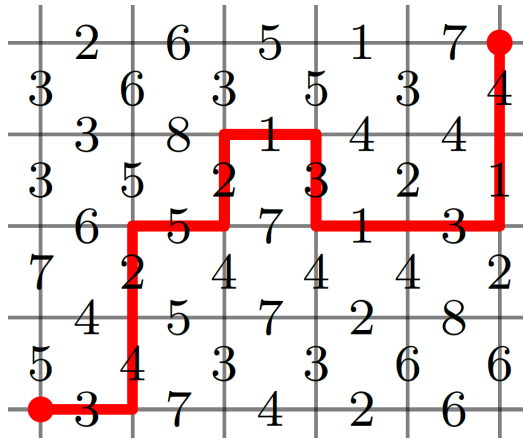
Growth process $B_t = \{v: \text{time from } (0,0) \text{ to } v < t\}$



Questions:

- ❖ Distance between two points:
Expectation? Fluctuation?
- ❖ What are the geodesics like?
- ❖ What does the boundary of the growth process look like?

First Passage Percolation (FPP)



✓ *Distance: Law of Large Numbers*

Limiting shape (Cox-Durrett, 81'):

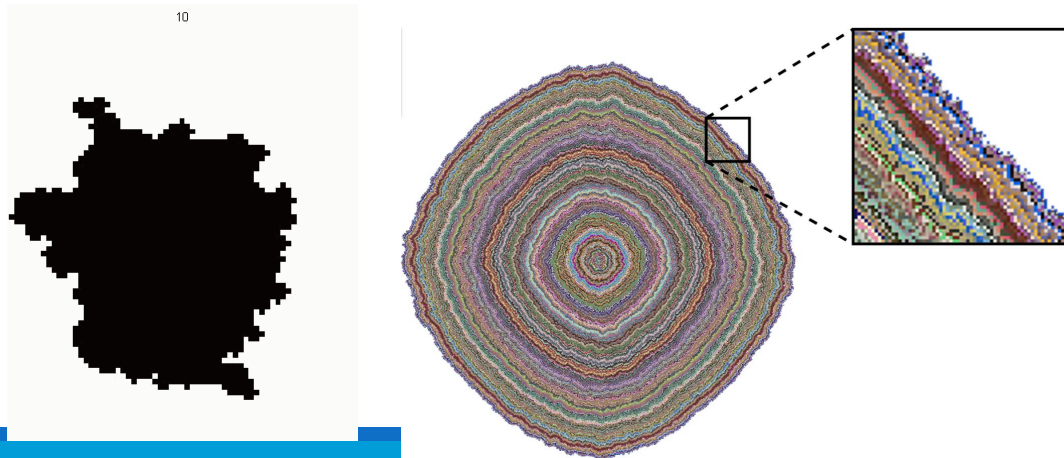
Assume the edge weight distribution μ satisfies:

- $\mu(\{0\}) < p_c$;
- $\mathbb{E}[\min\{\omega_1^2, \omega_2^2, \omega_3^2, \omega_4^2\}] < \infty$, for $\omega_1, \omega_2, \omega_3, \omega_4 \sim \mu$;

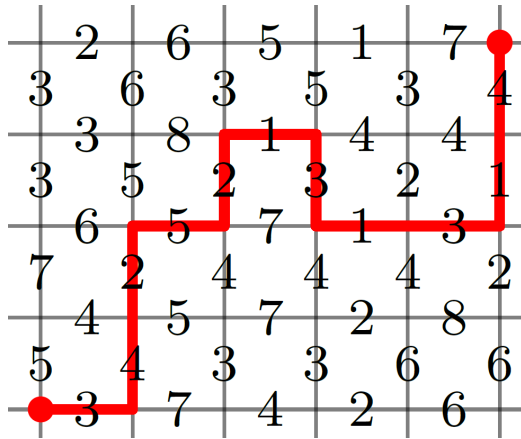
Then there is a deterministic, convex, compact $\mathcal{S}_\mu \subset \mathbb{R}^2$, such that almost surely,

$$(1 - \epsilon)\mathcal{S}_\mu \subset \frac{B_t}{t} \subset (1 + \epsilon)\mathcal{S}_\mu$$

for any $\epsilon > 0$, and large enough t .



First Passage Percolation (FPP)



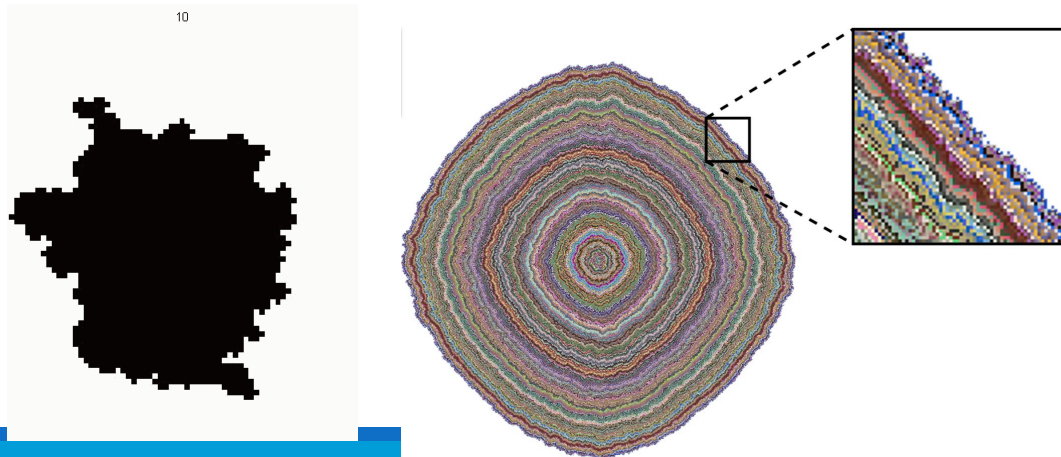
What is expected? (Strong KPZ universality conjecture)

For two points of Euclidean distance r

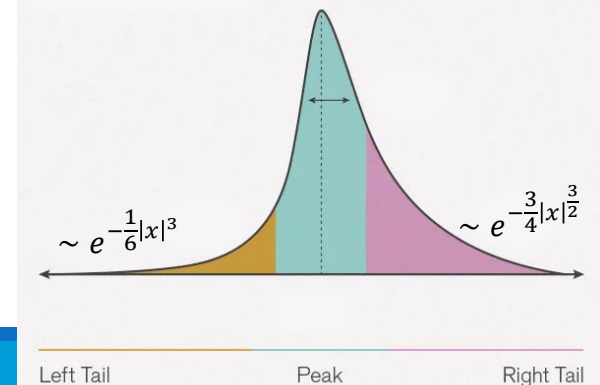
- FPP distance: order r (known by limiting shape)
- Fluctuation: order $r^{1/3}$

Fluctuation not Gaussian, but Tracy-Widom GUE

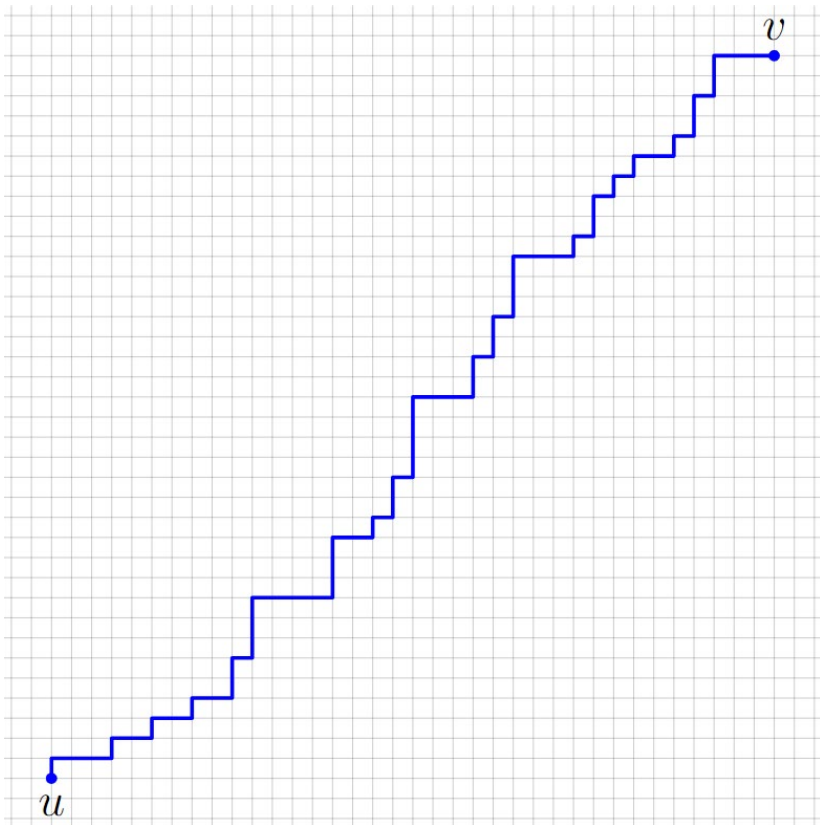
- Geodesic fluctuation from the straight line: order $r^{2/3}$
- Boundary of metric ball B_r : $r^{2/3} \times r^{1/3}$ scaling



TRACY-WIDOM DISTRIBUTION



Exponential Last Passage Percolation (Exp LPP)



- $\xi(v) \sim \text{Exp}(1)$, $\forall v \in \mathbb{Z}^2$ independently
($\mathbb{P}(\xi(v) > a) = e^{-a}$)
- Passage time: $T_{u,v} := \max_{\gamma} \sum_{w \in \gamma} \xi(w)$, over all up-right path γ
- Geodesic (max sum path): $\Gamma_{u,v} := \operatorname{argmax}_{\gamma} \sum_{w \in \gamma} \xi(w)$

Exact-solvable model: formulas exist

For example (Johannson, 00)

$$\mathbb{P}(T_{(1,1),(n,n)} < t) = \frac{1}{Z} \int_{[0,t]^n} \prod_{1 \leq i < j \leq n} (x_i - x_j)^2 \prod_{1 \leq i \leq n} e^{-x_i} dx_i$$

Using this, Johannson proves

$$\mathbb{P}(T_{(1,1),(n,n)} < 4n + x2^{4/3}n^{1/3}) \rightarrow F_{\text{TW}}(x)$$

i.e., $T_{(1,1),(n,n)} \approx 4n + 2^{4/3}n^{1/3}X$, with $X \sim \text{Tracy-Widom GUE}$

Directed landscape (DL)

Exponential Last Passage Percolation (Exp LPP)

- $\xi(v) \sim \text{Exp}(1), \forall v \in \mathbb{Z}^2$ independently
- Passage time: $T(u, v) := \max_{\gamma} \sum_{w \in \gamma} \xi(w)$, over all up-right path γ

(Dauvergne-Ortmann-Virag, 18', Dauvergne-Virag, 21')

As $n \rightarrow \infty$

$$(x, s; y, t) \mapsto 2^{-4/3} n^{-1/3} \left(T(R_n(x, s), R_n(y, t)) - 4n(t - s) - 2^{8/3} n^{2/3} (y - x) \right)$$

converges

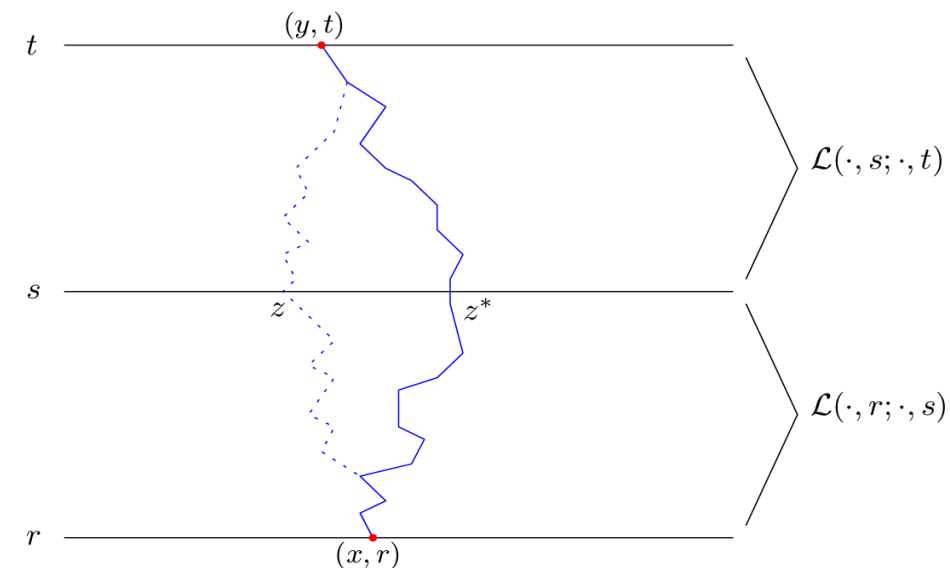
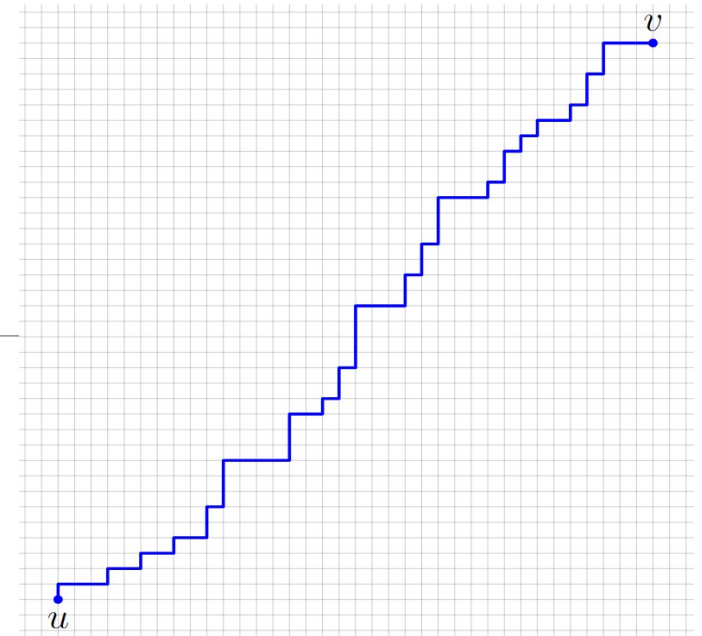
where $R_n(x, s) = (ns + 2^{5/3} n^{2/3} x, ns)$

The limit: a four-parameter random function

$$\mathcal{L} : \mathbb{R}_{\uparrow}^4 \rightarrow \mathbb{R} \quad \mathbb{R}_{\uparrow}^4 = \{(x, s; y, t) \in \mathbb{R}^4 : s < t\} \quad (\text{the directed landscape})$$

$$\text{Composition law} \quad \mathcal{L}(x, r; y, t) = \max_{z \in \mathbb{R}} \mathcal{L}(x, r; z, s) + \mathcal{L}(z, s; y, t)$$

Expect: scaling limit for FPP as well.

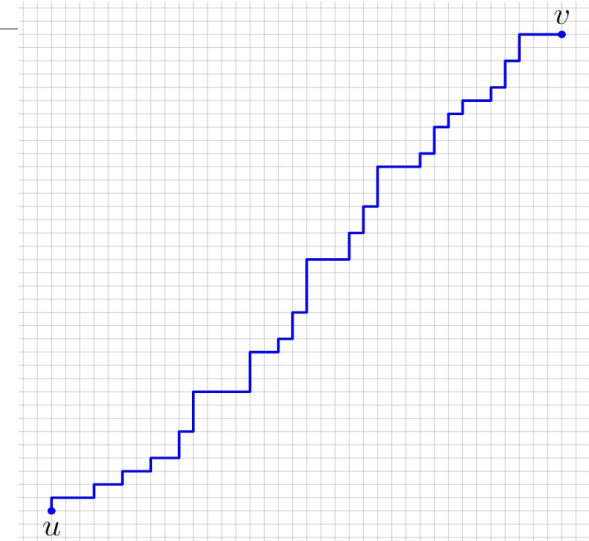
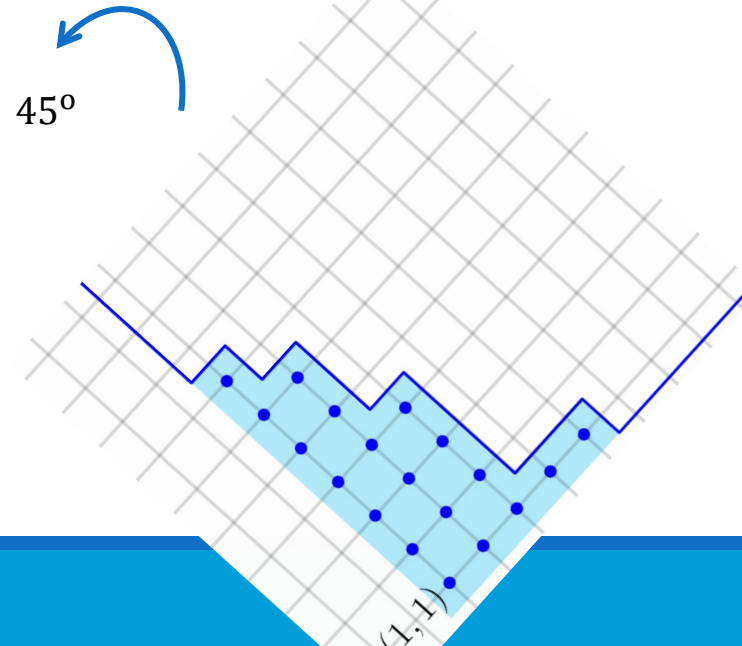
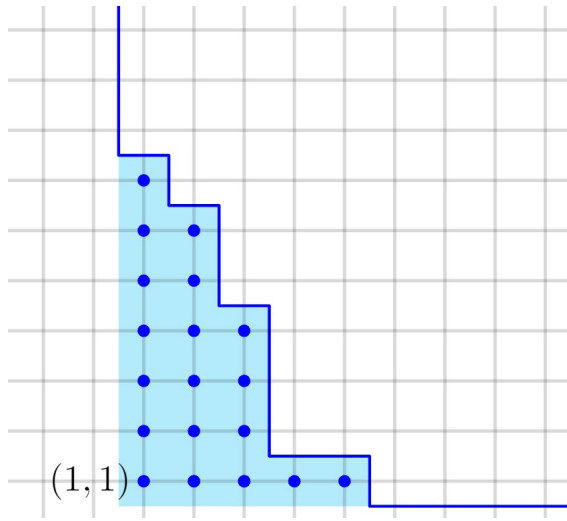


As a growth process

Exponential Last Passage Percolation (Exp LPP)

- $\xi(v) \sim \text{Exp}(1), \forall v \in \mathbb{Z}^2$ independently
- Passage time: $T(u, v) := \max_{\gamma} \sum_{w \in \gamma} \xi(w)$, over all up-right path γ

Metric ball $B_t = \{v \in \mathbb{N}^2 : T((1,1), v) < t\}$



Height function $H_t: \mathbb{Z} \rightarrow \mathbb{Z}$,
with $H_t(x+1) = H_t(x) \pm 1$,
and $H_0(x) = |x|$

- Markov process on the space of functions:
flip with rate 1 independently
- Can also take more general initial H_0

KPZ fixed point (KPZFP)

Height function process $(H_t)_{t \geq 0}$, starting from general H_0

For a family of them $(H_t^{(\varepsilon)})_{t \geq 0}$, scaling: $h_t^\varepsilon(x) = -\varepsilon^{1/2} \left(H_{2\varepsilon^{-3/2}t}^{(\varepsilon)}(2\varepsilon^{-1}x) - \varepsilon^{-3/2}t \right)$

(Matetski-Quastel-Remenik, 16')

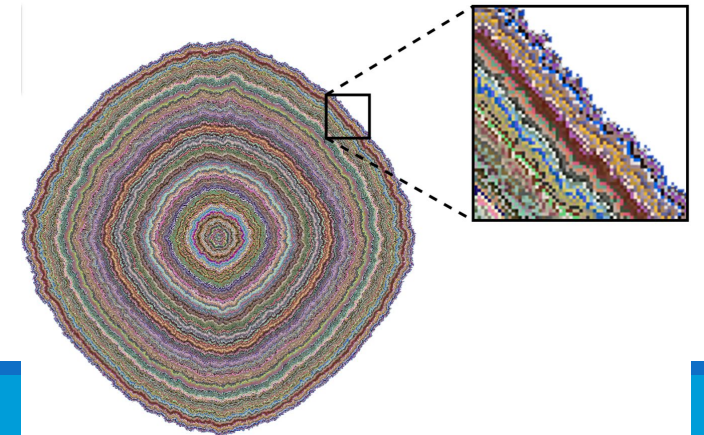
As $\varepsilon \rightarrow 0$, assume that h_0^ε converges to $f: \mathbb{R} \rightarrow \mathbb{R} \cup \{-\infty\}$. Then for $t > 0$, h_t^ε converges as $\varepsilon \rightarrow 0$, to $\mathfrak{h}_t f$, the **KPZ fixed point** starting from f at time t .

In words: KPZFP is a *Markov process on (upper semi-continuous) functions* $\mathbb{R} \rightarrow \mathbb{R} \cup \{-\infty\}$.

Exact formulas: for upper semi-continuous $f: \mathbb{R} \rightarrow \mathbb{R} \cup \{-\infty\}$ and lower semi-continuous $g: \mathbb{R} \rightarrow \mathbb{R} \cup \{\infty\}$, there is

$$\mathbb{P}(\mathfrak{h}_t f < g) = \det \left(\mathbf{I} - \mathbf{K}_{t/2}^{\text{hpyo}(f)} \mathbf{K}_{-t/2}^{\text{epi}(g)} \right)_{L^2(\mathbb{R})}$$

Expect: also for FPP



Random metric generates growth process

Growth process $(H_t)_{t \geq 0}$ starting from general H_0 : take S_0 according to H_0 ; define

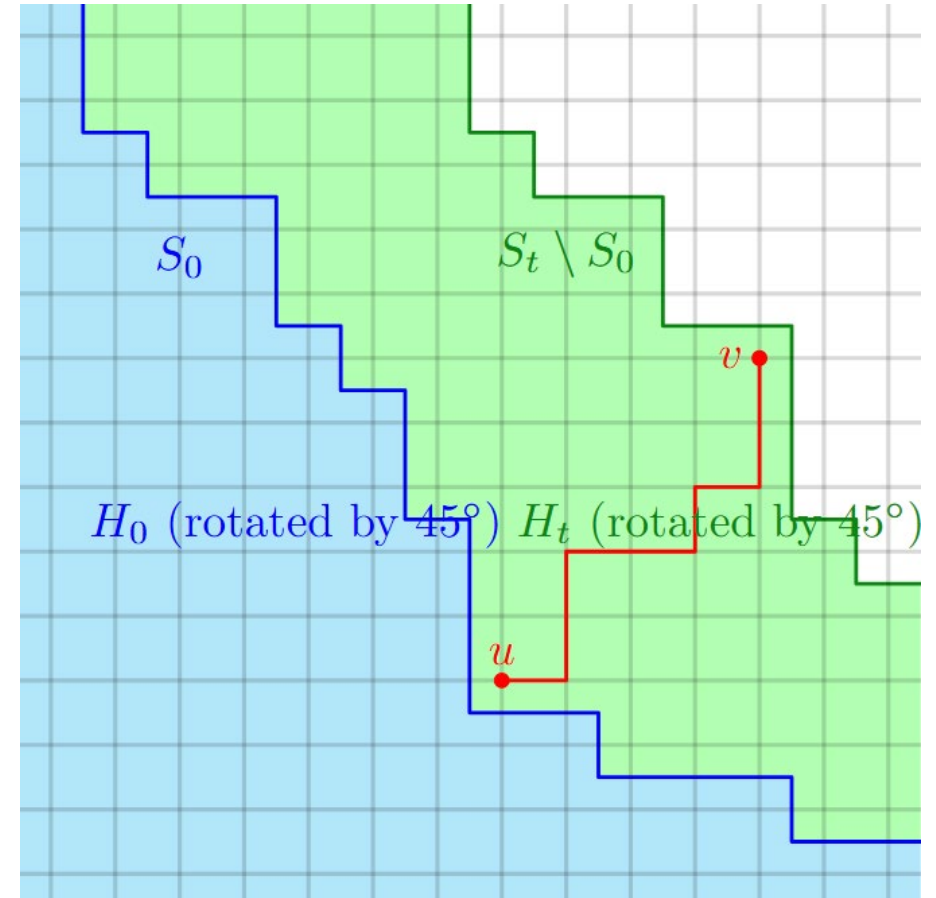
$$S_t = \{v \in \mathbb{Z}^2 : \max_{u \notin S_0} T(u, v) < t\}$$

Therefore, we have **variational formula**:

$$\mathfrak{h}_t f(y) = \max_x f(x) + \mathcal{L}(x, 0; y, t)$$

i.e., DL generates KPZFP. In general:

- KPZFP are *lower dimensional projections* of DL
- Random metric converges to DL $\Rightarrow$
Corresponding growth process converges to KPZFP

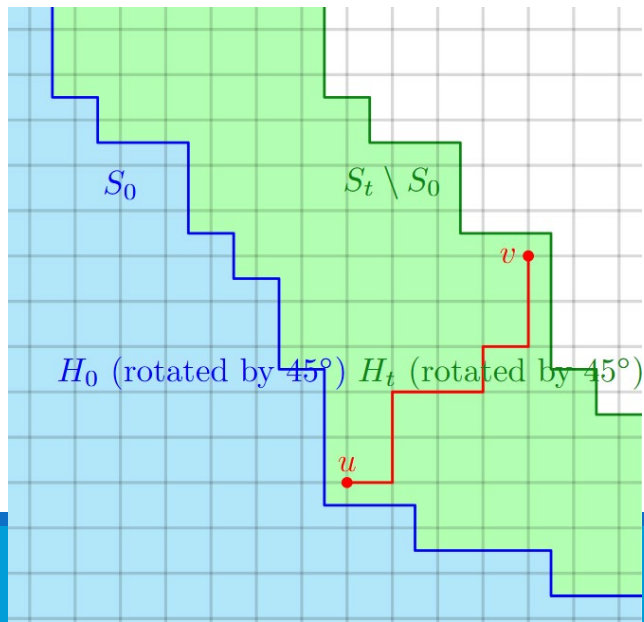


The opposite direction?

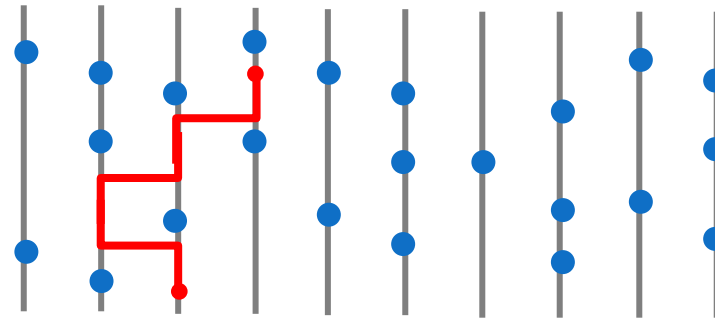
- Is DL “determined by” KPZFP?
- Growth process converges to KPZFP $\Rightarrow$ (certain) random metric converges to DL?

A motivation KPZFP convergence might be easier, in some cases.

In general, this convergence **upgrade** is not so canonical: the same growth process can be generated by *different* random metrics.



Example Growth process from Exp LPP, also from a Poisson field FPP



The opposite direction?

Characterize DL from KPZFP:

Dauvergne, Z., 2024 There is a unique (in law) $\mathcal{M}: \{(x, s; y, t): s < t\} \rightarrow \mathbb{R}$ that is continuous, and

1. For any $s < t$ and $f: \mathbb{R} \rightarrow \mathbb{R} \cup \{-\infty\}$, supported on finitely many points, we have (in law)

$$\sup_x \mathcal{M}(x, s; \cdot, t) + f(x) = \mathfrak{h}_{t-s} f,$$

2. For any $r < s < t$, and x, y, z , we have $\mathcal{M}(x, r; y, t) \geq \mathcal{M}(x, r; z, s) + \mathcal{M}(z, s; y, t)$,

3. For any disjoint intervals $\{(s_i, t_i)\}_{i=1}^k$, $\mathcal{M}(\cdot, s_i; \cdot, t_i)$ are independent.

Note:

❖ This relies on the construction of DL; i.e., we show that $\mathcal{M} = \mathcal{L}$ in distribution.

❖ This can also be done **without relying on** the construction of DL, if there is one more condition:

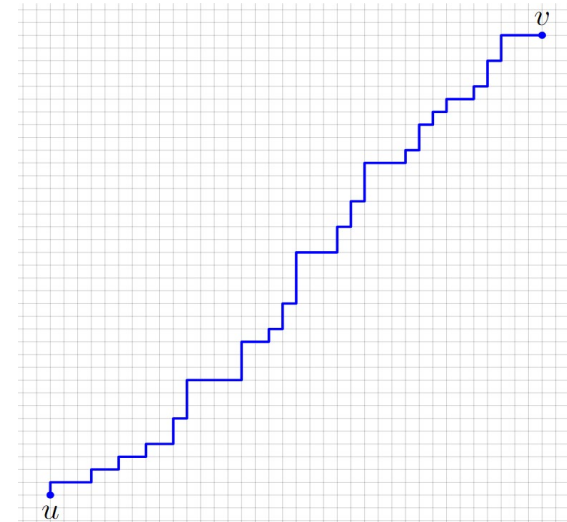
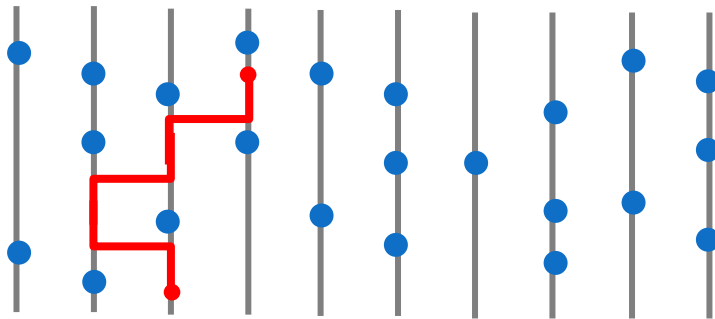
4. For the stationary horizon $\{H^{(i)}\}_{i=1}^k$, we have $\left\{ \sup_x \mathcal{M}(x, s; \cdot, t) + H^{(i)}(x) + C_i \right\}_{i=1}^k = \{H^{(i)}\}_{i=1}^k$ in law.

This gives an alternative (from *Dauvergne-Ortman-Virag, 16'*) and less integrable construction of DL, from KPZFP.

Applications to DL convergence

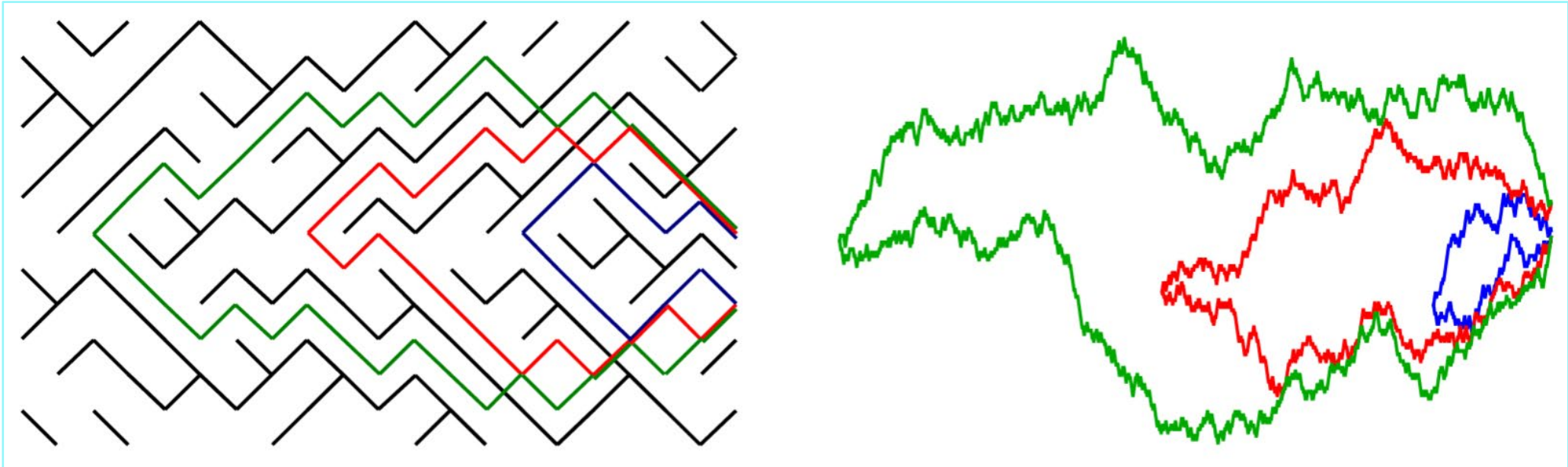
To prove DL convergence given KPZFP convergence:
check 2 & 3 for any subsequential limit (of the random metrics).

e.g., from growth process convergence to KPZFP in *Matetski-Quastel-Remenik, 16'*,
both Exp LPP and Poisson field FPP converge to DL



In summary: KPZ FP convergence is now *equivalent* to DL convergence

Applications to DL convergence



Brownian web/Coalescing rand walk distance

- For generated random growth, KPZFP convergence shown (*Veto-Virag, 23'*) using a bijection to Seppalainen–Johansson model or Brownian LPP
- No bijection as a random metric

Proof ideas & insights

Heuristics (of why this characterization holds?)

- DL is a random function on $\mathbb{R}_\uparrow^4 = \{(x, s; y, t): s < t\}$, while KPZFP $\mathfrak{h}_t f$ is on $\mathbb{R}$
 - Seems that DL has **much more** information?
- Not really! Because geodesics coalesce.

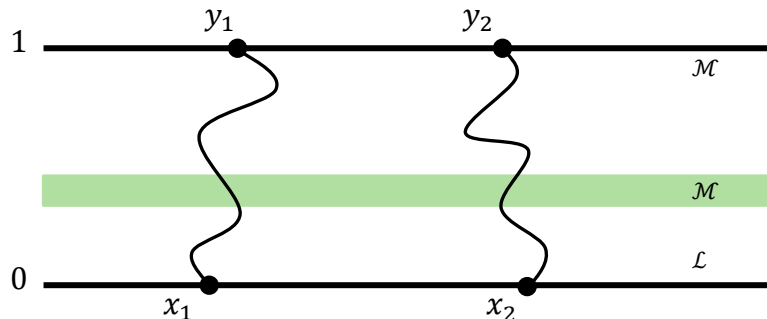
In DL, for $|x - x'|$ and $|y - y'|$ small, likely

$$\mathcal{L}(x, 0; y, 1) + \mathcal{L}(x', 0; y', 1) = \mathcal{L}(x, 0; y', 1) + \mathcal{L}(x', 0; y, 1)$$

Proof to show equality in law $(\mathcal{M}(x_1, 0; y_1, 1), \mathcal{M}(x_2, 0; y_2, 1)) = (\mathcal{L}(x_1, 0; y_1, 1), \mathcal{L}(x_2, 0; y_2, 1))$

In practice, use a *Lindeberg exchange strategy*, plus careful quantitative analysis, to exploit **coalescence**

(Originated in Lindeberg's proof of CLT, 1922; used in e.g., hydrodynamics *Bohadoran-Guiol-Ravishankar-Saada*, 02'; spin glass *Chatterjee*, 04'; random matrices: *Chatterjee*, 05', *Tao-Vu*, 07', *Knowles-Yin*, 17'; (2+1)D growth, *Caravenna-Sun-Zygouras*, 21', *Tsai*, 24')



Equal with high probability
(under a certain coupling)

